Algebra Review: Unit 7 — Polynomials & Factoring

Answer Key

Station 1: Adding and Subtracting Polynomials

1. $-2x^3 + 5x + 4$; degree 3; trinomial
2. Degree 4; binomial
3. $4x^2 - 3x + 4$
4. $5x^2 - 5x + 6$
5. $5x^3 - x^2 + 2x - 6$
6. $7a^2 - 4a + 7$
7. $5x^2y - 2xy$
8. $-2x^2 + 9$
9. $x^2 + 5x + 2$
10. $6y^2 - 6y + 11$

Station 2: Multiplying Polynomials

1. $12x^3 - 15x$
2. $-2a^3 - 6a^2 + 8a$
3. $x^2 + 11x + 28$
4. $x^2 - 2x - 15$
5. $6x^2 - 5x - 4$
6. $x^2 + 12x + 36$
7. $9x^2 - 12x + 4$
8. $x^2 - 64$
9. $x^3 - x^2 - x + 10$
10. $2x^2 - x - 15$

Station 3: Factoring: GCF and Trinomials with $a = 1$

1. $6\,(x + 3)$
2. $4x\,(3x - 2)$
3. $5a^2b\,(3a - 2b)$
4. $(x + 3)\,(x + 4)$
5. $(x - 5)\,(x - 4)$
6. $(x - 4)\,(x + 6)$
7. $(x - 7)\,(x + 2)$
8. $(x - 7)\,(x - 4)$
9. $3\,(x + 3)\,(x + 4)$
10. $(x - 5)\,(x + 8)$

Station 4: Factoring Special Products and Trinomials with $a \neq 1$

1. $(x - 7)\,(x + 7)$
2. $(2x - 5)\,(2x + 5)$
3. $(x + 5)^2$
4. $(3x - 2)^2$
5. $(x + 3)\,(2x + 1)$
6. $(x - 2)\,(3x - 4)$
7. $(2x + 3)\,(3x - 4)$
8. $5\,(x - 3)\,(x + 3)$
9. $2x\,(x - 3)\,(x + 3)$
10. $4\,(x - 2)\,(x + 3)$